

Workshop 5

Problem 1

Keywords: Limit Definition of Derivative, Derivative Rules

Evaluate each of the following limits **using the methods from Section 3.1**. This means you should identify a function $f(x)$ and a point a such that the limit in the problem is equivalent to computing $f'(a)$. Then, use derivative rules from §3.3 to compute each derivative, and by doing so, compute each limit. (Note: You may not use differentiation rules from §3.6 in this exercise.)

- (a) $\lim_{x \rightarrow 3} \frac{(x^2 + 3x + 2) - 20}{x - 3}$. Check your answer to this problem using methods from Chapter 2.
- (b) $\lim_{h \rightarrow 0} \frac{e^{2+h} - e^2}{h}$
- (c) $\lim_{y \rightarrow 2} \frac{y^5 + y^4 - 4y^3 - 3y^2 - 4}{y - 2}$
- (d) $\lim_{x \rightarrow 4} \frac{\sqrt{x} - x + 2}{x - 4}$.

Problem 2

Keywords: Derivative as a Slope, Tangent Lines

Consider two quadratic functions $f_1(x) = x^2$ and $f_2(x) = C - x^2$, where C is an unspecified positive constant. We say that two graphs are *orthogonal* if their tangent lines at points of intersection are perpendicular.

- (a) Find the points of intersection of the graphs of f_1 and f_2 .
- (b) Find the values of the derivatives of f_1 and f_2 at the points of intersection.
- (c) Find the value of C so that the graphs of f_1 and f_2 are orthogonal.
- (d) Find the values of the constant D so that the parabolas $y = x^2$ and $y = (D - x)^2$ are orthogonal.
- (e) Find the value of the positive constant E so that the parabola $y = Ex^2$ and the curve $y = \frac{1}{x^2}$ are orthogonal.

Problem 3

Keywords: Tangent Lines, Derivatives as Slopes

- (a) Find all the points on the curve $y = x^3 + x^2 - x$, if any, where the tangent line is horizontal.
- (b) Find all the points on the curve $y = x^3 + x^2 + 2x$, if any, where the tangent line is horizontal.
- (c) For which values of b does the curve $y = x^3 + x^2 + bx$ have 2 points where the tangent is horizontal? 1 point? 0 point? Check that this is consistent with your answers to the previous 2 questions. Can there be more than 2 points where the tangent is horizontal?
- (d) Let f be a differentiable function. If $y = f(x)$ has a horizontal tangent at $x = c$, does $y = f(x)^2$ also have a horizontal tangent at $x = c$? What about $y = xf(x)$? Justify your answer.